

Reinforcement Learning

Task 1

Allwin Suresh BL.SC.P2DSC25005

Part A: Explore Reinforcement Learning Applications

Identify and list any five real-world applications where Reinforcement Learning is used.

1. Autonomous Driving
 - RL helps self-driving cars make real-time decisions such as lane changing, obstacle avoidance, adaptive cruise control, and traffic navigation by learning from interactions with the environment.
2. Robotics and Industrial Automation
 - Robots use RL to learn tasks like object manipulation, assembly, warehouse automation, and navigation without requiring explicit programming for every scenario.
3. Healthcare
 - RL is applied in personalised treatment planning, drug dosage optimisation, medical diagnosis support, and patient monitoring by learning treatment policies.
4. Recommendation Systems
 - Streaming and e-commerce platforms such as Netflix, YouTube, Spotify, and Amazon use RL to personalise content and product recommendations based on user interactions and feedback.
5. Smart Grid and Energy Management
 - RL is used to optimise electricity distribution, renewable energy integration, battery charging/discharging, and HVAC control in smart buildings to reduce energy consumption and operational costs.

Part B: Research Paper Exploration

Select any two applications from Part A.

For each application, identify one recent research paper (published within the last 3 years is preferred) and summarise it in the following form

Item	Paper - 1	Paper - 2
Application	Autonomous Driving	Healthcare (Sepsis Treatment)
Paper Title	HCRMP: A LLM-Hinted Contextual Reinforcement Learning Framework for Autonomous Driving	Off by a beat: the effects of temporal misalignment in reinforcement learning for sepsis treatment
Authors	Zhiwen Chen, Bo Leng, Zhuoren Li, Hanming Deng, Guizhe Jin, Ran Yu, Huanxi Wen	Shengpu Tang, Jiayu Yao, Jenna Wiens, Sonali Parbhoo
Year	2025	2026
Journal / Conference	arXiv (cs.RO)	npj Digital Medicine (Nature Portfolio)

Item	Paper - 1	Paper - 2
Main Contribution	Proposes HCRMP, an LLM-hinted contextual reinforcement learning framework for autonomous driving. Instead of allowing the LLM to directly control driving decisions, it provides semantic hints to the RL agent, reducing the impact of LLM hallucinations. Experiments in the CARLA simulator achieved up to 80.3% task success rate and reduced collision rates by 11.4% in safety-critical driving scenarios.	Demonstrates that temporal misalignment in healthcare RL datasets leads to incorrect treatment recommendations. The paper introduces a shifted temporal alignment strategy that preserves causality and improves the reliability of RL-based sepsis treatment policies, preventing misleading recommendations and improving patient safety.
RL Algorithm	Multi-Critic Proximal Policy Optimization (PPO) integrated with LLM-guided semantic hints and Retrieval-Augmented Generation (RAG).	Reinforcement Learning for Markov Decision Process (MDP) formulation of sepsis treatment using Offline RL policy learning and off-policy evaluation. The paper focuses on correcting temporal alignment rather than proposing a new RL algorithm.